

Learning Relational Action Rules Without a Model

*Inducing first-order rules from demonstrations,
and a causal ablation of the learned competence*

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ABSTRACT

We study whether a symbolic learner can acquire general relational action rules purely from labeled demonstrations (before/action/after state triples) with no language model consulted at any point, and whether the resulting competence is *causally* carried by the learned rules. From a small demonstration set the learner induces two first-order rules with variables: a static derivation rule and a state-transition rule with an explicit deletion effect. Learning is verified model-free by an enforced policy that blocks and counts any model call (measured: zero throughout). On a frozen held-out evaluation of 14 cases containing only unseen constants (positive derivations, missing-precondition negatives, role-reversal traps, irrelevant-fact distractors, and multi-step transition chains) the fully-equipped system solves 14/14. We then ablate the learned state across seven conditions, each in its own isolated clone. Removing the learned rules, or leaving them present but denying them applicable status, drops the score to 7/14 (exactly the cases that require no derivation) while a sham clone-and-restore control and a semantic-store deletion control both retain 14/14, and reinstating the deleted rules recovers 14/14. The competence is thus localized to the induced, validated rules and to nothing else. We then report what the same procedure costs and what it does when its evidence does not settle the matter: one demonstration is refused as a case rather than a generalisation and three suffice; the counter-demonstrations are what tighten the rule body, which can be read before the rule is trusted; and where several hypotheses survive the evidence, concluding only what all of them accept yields **zero** wrong conclusions where concluding on any one of them yields 26, while the learner states the case that would decide the matter, closing **16 of 16** such ambiguities against 1 of 16 for randomly chosen further examples.

1 Rule induction, and why ablation

Much current generalization relies on large pretrained models whose behavior is hard to attribute. We ask whether general, variable-carrying relational rules can be learned directly from a handful of examples, verifiably without any model in the loop: the classical goal of inductive logic programming [1,2,3]. A learned rule must generalize over *argument structure*, not surface co-occurrence: it must fire on entirely unseen constants, ignore irrelevant facts, refuse when a precondition is missing, and refuse when relations hold but point the wrong way (role reversal). It must also express *deletion* effects, so that applying a transition rule changes the world state rather than merely accumulating conclusions [4].

General relational action rules can be induced from a handful of demonstrations without any language model, and the acquired competence is causally attributable to the learned rules: a controlled ablation removes the capability by deleting the rules (or revoking their applicable status) and restores it by reinstating them, while unrelated state and the cloning procedure itself are shown to be non-causal.

A system can pass a benchmark for the wrong reason. Demonstrating that a capability is *present* is not the same as showing *what carries it*. A causal ablation (remove the hypothesized carrier and watch the capability vanish, restore it and watch it return, with code, runtime, inputs, and model-availability held fixed) converts a performance claim into a mechanistic one.

2 Method

Demonstrations as evidence. A teacher supplies labeled examples only: each is a before-state, an optional action, and an after-state, tagged positive or negative. The teacher asserts no rule. Negative examples hold all-but-one precondition and show the effect *not* occurring, forcing each precondition into the rule body rather than leaving it as an incidental fact.

Generalization by anti-unification. From the positive examples the learner computes a least-general generalization, replacing constants with typed variables shared consistently across argument positions [5], and keeps only the body literals discriminated by the negatives. This yields a first-order rule with add-effects and, where the after-state drops a fact, an explicit delete-effect.

Validation gate. An induced rule becomes *applicable* only after it is confirmed on separate validation demonstrations it never saw during induction. Rules that are present but not validated remain non-applicable.

Evaluation by application only. A frozen suite of unseen cases is answered purely by applying already-persisted applicable rules: never by inducing a new one. Two policies are asserted before any component loads: no learned model may be consulted (attempts are counted and asserted zero), and during evaluation no rule may be induced or re-validated. The frozen policy is what makes a *negative* ablation result meaningful: an ablated system cannot silently re-derive the capability it was stripped of.

Ablation design. The hypothesized carrier (the learned rules) is removed, degraded, or restored across seven conditions, each run against its own cloned copy in its own process so no cached state leaks. A sham condition clones and restores without ablating (controlling for the cloning procedure); a semantic-store-deletion condition is a negative control for a component that should not matter.

3 The learned rules

Two rules were induced (relation symbols are abstract; roles are what matter). A static derivation rule concludes a relation when four conditions on its two arguments hold. A transition rule fires on an action and its preconditions, adds the resulting location fact, and (the delete effect) retracts the origin fact, so applying it moves the world forward:

Rule	Induced from	Form (schematic)
Static derivation	6 demos (2 pos, 4 neg)	$R_1(x,y) \wedge P(x) \wedge Q(y) \wedge R_2(x,y) \rightarrow S(x,y)$
State transition	5 demos	$\text{Move}(x,b,a) \wedge \text{Open}(a) \wedge \text{Path}(b,a) \rightarrow \text{At}(x,a) \oplus \text{At}(x,b)$

Table 1. The two induced rules, both confirmed on held-out validation demonstrations. $\oplus$ marks the delete-effect (the origin fact is retracted on application).

4 Results

On the frozen suite of 14 cases (all using unseen constants: positive derivations, missing-precondition negatives, an irrelevant-fact distractor, role-reversal traps, and transition-sequence cases) the fully-equipped system solves 14/14 with zero model calls.

A causal ablation of the learned competence

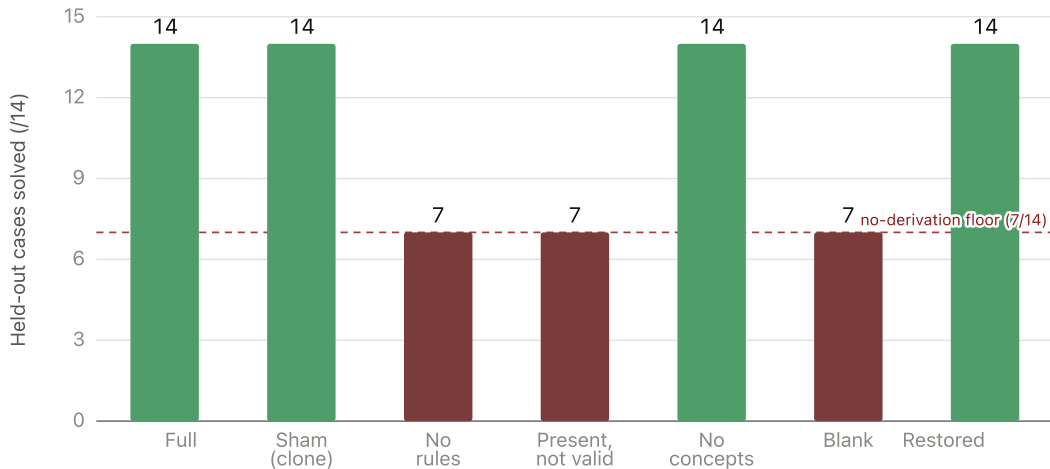


Figure 1. Held-out cases solved (out of 14) across seven conditions. Green retains full competence; red collapses to the 7/14 floor. Deleting the learned rules (No rules) or revoking their applicable status (Present, not valid) removes exactly the derivation-requiring cases; deleting the semantic concept store (No concepts) changes nothing; the sham clone-and-restore matches Full; and reinstating the rules (Restored) recovers 14/14.

Condition	Rules avail.	Applicable	Solved	Model calls
Full	2	2	14/14	0
Sham (clone + restore)	2	2	14/14	0
No learned rules	0	0	7/14	0
Present, not validated	2	0	7/14	0
No concepts (control)	2	2	14/14	0
Blank state	0	0	7/14	0
Restored	2	2	14/14	0

Table 2. The seven ablation conditions. Every condition ran with zero model attempts under an enforced model-free policy. “Present, not valid” isolates the validation gate itself: the rules exist but none is applicable, so the score is the same as deleting them.

The 7/14 floor is not noise: the seven cases that pass under every ablated condition are exactly the seven that require *no* derivation (missing-precondition refusals, role-reversal refusals, a blocked destination). The seven that flip from pass to fail when the rules are removed or de-validated are exactly the seven requiring an actual derivation. This is the clean causal signature: removing the rules removes precisely the positive-derivation competence and nothing else.

Remove the rules, the capability goes; restore them, it returns



Figure 2. Reversibility: full competence (14/14), rules removed (7/14), rules reinstated (14/14). Because inputs, code, runtime, and model-availability are held fixed across the three, the learned rules are the difference that makes the difference.

5 What the same learner takes, and what it does when the evidence does not decide

The result above is a single teaching set carried to a single conclusion. Two questions it does not answer are how much evidence the procedure needs, and what it does when the evidence it has admits more than one hypothesis. Both have since been measured on the same inducer, applied to classification over features a

perceptual faculty measured off real images rather than to action triples supplied by a teacher. The full study and its data are reported separately [6]; what follows is what bears on the claims made here.

5.1 Re-verification

The frozen suite was re-run against the live substrate on 18 September 2026 and solved **14/14**, applying the persisted rules under the same frozen learning policy. The model-free condition is now stronger than the one originally reported: rather than a policy that blocked model calls and counted them, the model entry points have been removed from the system altogether, and the condition asserts that none of them can be imported. There is no longer a model to consult, so there is nothing to block.

5.2 How much evidence a rule takes

Holding the counter-demonstrations fixed and varying the positive demonstrations from one to four (Figure 3): with a *single* positive the learner declines to generalise at all and says why, that one example is a case rather than a generalisation. With two the evidence does not yet determine a single hypothesis. From **three** the induced rule applies to every held-out case, and a fourth changes nothing, which is the same property this paper reports for the transition rule in section 6: a further success cannot narrow a hypothesis that already covers it. At every setting, refusals on non-members stayed at 100%: thinner evidence buys less application, never more error.

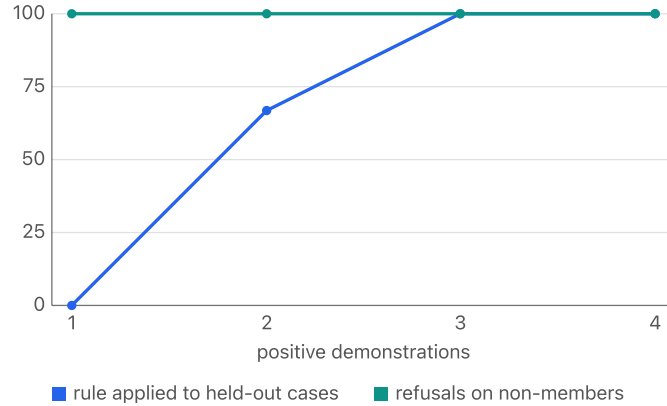


Figure 3. Outcome against the number of positive demonstrations. One demonstration is refused as a case rather than a generalisation; three are enough for the rule to apply to every held-out case. Refusals on non-members are 100% at every setting, so the cost of thin evidence is silence rather than a wrong conclusion.

5.3 What the counter-demonstrations buy

This paper argues that negatives are what force a precondition into the body rather than leaving it an incidental fact. That argument can now be read off the induced rule itself. Holding the positives fixed and varying the counter-demonstrations from none to two (Figure 4): with none, nothing separates the hypothesis from its background and no rule is induced. With one, the body is a single literal, which is exactly what that evidence licenses, and the rule accordingly applies to a case it should refuse. With the second, which dis-

criminate the remaining feature, the body becomes the full conjunction, application is complete and the refusals are restored.

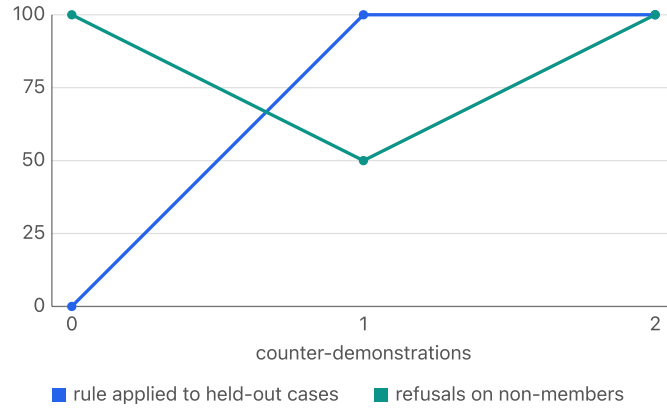


Figure 4. Outcome against the number of counter-demonstrations. Application rises with the first counter-demonstration; correct refusal is restored by the second, which is the one that discriminates the remaining precondition. The induced body at each setting is exactly as general as the evidence supports, and can be read before the rule is trusted.

5.4 When the evidence admits more than one hypothesis

A learner that keeps one rule must choose whenever its evidence admits several, and a choice the evidence does not support is a guess with a rule’s authority. This learner keeps every hypothesis consistent with what it was shown, which moves the question to what *applying* them should do. Over 32 inductions and 416 images, teaching sets whose positives shared an incidental feature left two hypotheses standing in every case, and teaching sets whose positives varied left one in every case. Scored over the same 288 held-out instances, three policies over those recorded hypotheses differ sharply (Table 3).

Policy over the surviving hypotheses	Applied to members	Applied to non-members	Wrong conclusions
Conclude if any surviving hypothesis fires	100% (96/96)	13.5% (26/192)	26
Conclude only where one hypothesis survived	50.0% (48/96)	0% (0/192)	0
Conclude only what every hypothesis accepts	72.9% (70/96)	0% (0/192)	0

Table 3. Three policies scored on identical recorded hypotheses. Requiring unanimity recovers most of the application that waiting for a determined rule gives up, and costs nothing in wrong conclusions, because the instances the hypotheses disagree about are exactly the instances the evidence does not decide.

An undetermined induction also reports, in its own terms, the case that would eliminate one of its hypotheses: an instance that is one thing but not another, which one hypothesis accepts and the other refuses. Supplying the case it asks for closed the ambiguity in **sixteen of sixteen** such inductions, in a mean of 2.6 rounds, where supplying randomly chosen further examples under the same budget closed **one** of sixteen. This is the counter-demonstration discipline of section 2 turned around: instead of a teacher knowing which negative to supply, the learner states which case it needs.

5.5 The same procedure over perceived structure

Nothing in the induction is specific to demonstrations a teacher writes down. In a separate study the same procedure was given individuals admitted by a perceptual faculty, each carrying the colour, shape and size measured off a real image, with no features supplied by hand. Over thirty images every measured thing was admitted as an individual holding every feature measured of it, and categories induced from four such individuals each applied correctly to held-out ones, with **zero** wrong conclusions and complete refusal on things of categories never taught. A rule induced over what a teacher demonstrated and a rule induced over what a faculty perceived are the same kind of object, learned by the same procedure and applied by the same reasoner.

6 Limitations and conclusion

The transition rule reported in every measured result was induced from five demonstrations, and its body omitted an explicit source-location precondition. We have since re-run that case, and the resolution is a direct instance of the paper's own claim about what constrains a rule body. Adding a sixth *positive* demonstration leaves the induced body unchanged, $\text{PATH}(\text{?X2}, \text{?X1}) \wedge \text{OPEN}(\text{?X1}) \wedge \text{MOVE}(\text{?X0}, \text{?X2}, \text{?X1})$, because a further success cannot narrow a hypothesis that already covers it. A counter-demonstration does: one case in which the mover is not at the source and the action therefore has no effect forces $\text{AT}(\text{?X0}, \text{?X2})$ into the body, yielding the fully constrained rule $\text{AT}(\text{?X0}, \text{?X2}) \wedge \text{PATH}(\text{?X2}, \text{?X1}) \wedge \text{OPEN}(\text{?X1}) \wedge \text{MOVE}(\text{?X0}, \text{?X2}, \text{?X1}) \rightarrow \text{AT}(\text{?X0}, \text{?X1})$. The measured gap was thus a property of the teaching set, not of the learner, and it is closed by the same discipline (counter-demonstrations constrain) that the relational result rests on. The relations are synthetic nonce symbols chosen to rule out lexical priors; the suites are fixed and modest in size. Within that scope the result is strong and, we think, unusually clean for a learning claim: two general relational rules are induced from a handful of demonstrations with no model in the loop, and a controlled ablation localizes the resulting competence to those rules (deleted, it vanishes; restored, it returns) while the concept store and the cloning procedure are shown to carry none of it.

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